

Question 1 (5 points)

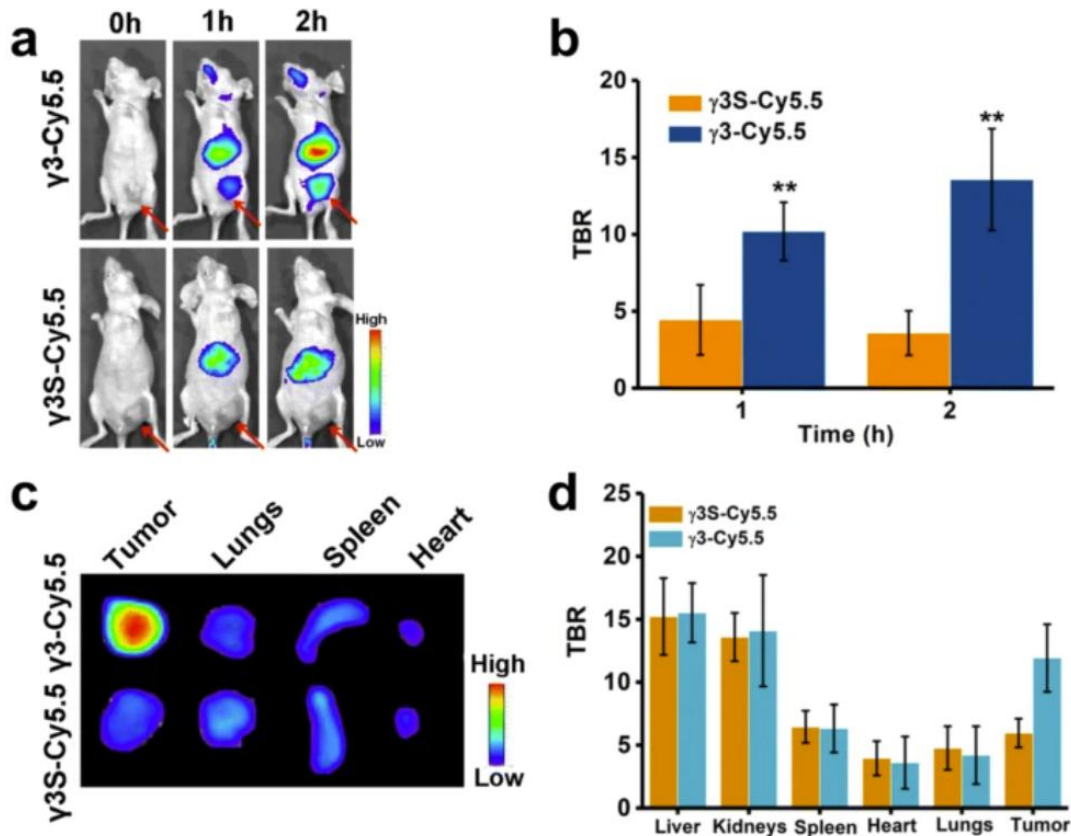


Figure 1 **a** In vivo NIRF of mice with subcutaneous MDA-MB-231 xenografts receiving $\gamma 3$ -Cy5.5 and $\gamma 3S$ -Cy5.5. Mice were imaged at 0, 1, and 2 h after injection. **b** The TBR observed on the in vivo NIRF images. **c** Ex vivo NIRF imaging and **d** TBR in the tumors, lungs, spleens, and hearts after i.v. administration of $\gamma 3$ -Cy5.5 and $\gamma 3S$ -Cy5.5 into MDA-MB-231-bearing mice for 2 h. ($n = 3$; $**P < 0.01$). (From Zhang et al., *Molecular Imaging and Biology*, 2019). MDA-MB-231, TNBC cell line; TBR: tumor to background ratio, background signal was measured in the ear.

Triple negative breast cancer (TNBC) is generally associated with poor patient prognosis as there are limited therapeutic options available for this type of cancer, in particular in patients with advanced disease. This makes novel diagnostic tools for early disease detection urgent.

In a recent study, intercellular adhesion molecule 1 (ICAM-1) was identified as novel marker overexpressed by TNBC cells. A 17 amino-acid-long peptide called $\gamma 3$ was found to specifically bind to the ICAM-1 on tumor cells. In order to evaluate the feasibility of utilizing $\gamma 3$ imaging for the detection of TNBC an optical fluorescence imaging experiment was performed. To this end the peptide $\gamma 3$ was labeled with the fluorescence dye C5.5 (ex. 678 nm, em. 694 nm) and imaged using fluorescence reflectance imaging. A scrambled control peptide $\gamma 3S$ was used as non-binding control.

Please analyze the results from this experiment shown in Figure 1 and answer the following questions:

- 1) Which hardware components are essential for the functionality of the imaging device used to produce the images shown in Fig. 1a, c ? (1 point)
- 2) Please explain how the images shown in Fig. 1a and Fig. 1c were generated. (1 point)
- 3) Why was Cy5.5 chosen as fluorescent dye to label the peptide? Could you suggest another fluorescent dye suitable for this experiment? (1 point)
- 4) Based on the results shown in Figure 1: can the newly identified peptide $\gamma 3$ be used for the detection of TNBC? Please justify your answer in a few sentences. (1 point)
- 5) Is optical imaging the appropriate imaging readout for the setup of a diagnostic tool to be used clinically? Why or why not? (1 point)

1) NIR-laser, CCD Camera, Fluorescence filter

2) Antibodies were directly injected, left to spread and later excited using the NIR and fluorescence captured on camera. For picture c the organs were isolated for imaging

3) As it does not quickly degrade, low autofluorescence and low absorption by the tissue, Cy7

4) It's on a good way but still spreads quite a bit on other organs. It could use a higher specificity. However the difference between the control and actual antibody on the tumor

5) Optical Imaging can bei Menschlichen Tissue nicht genug einsehen